

Inverting Buck-Boost

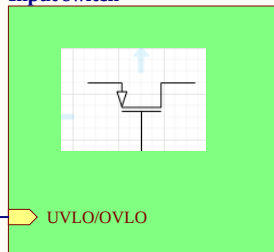


J1
108-0906-001
+24V

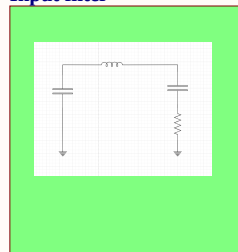


J4
108-0903-001
GND

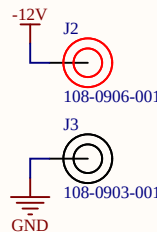
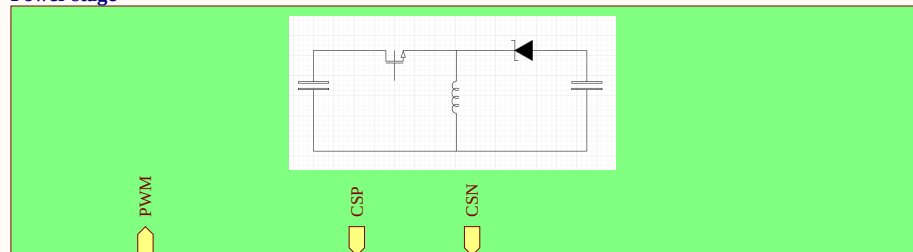
Input switch



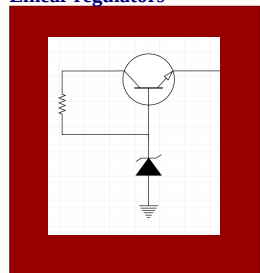
Input filter



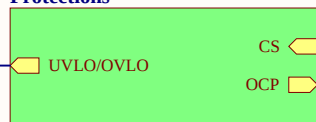
Power stage



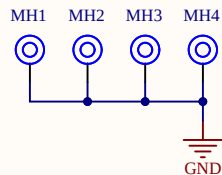
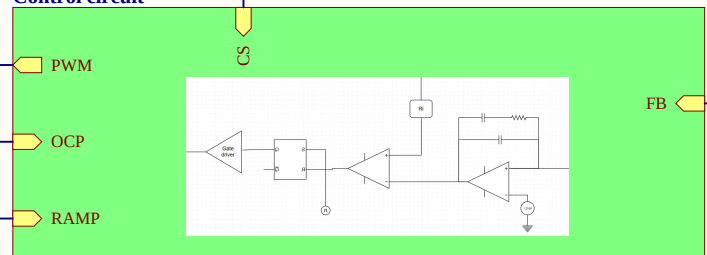
Linear regulators



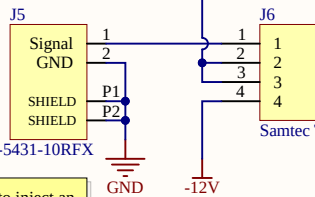
Protections



Control circuit



J5
Signal
GND
SHIELD
SHIELD
031-5431-10RFX



Samtec TS-104-T-AA

J6 is a 4-pin header that selects the op-amp negative input source: either the injected sawtooth signal (to subtract it from the compensator output) or -12 V to disable this feature.

J5 is a female BNC connector used to inject an external compensating ramp (sawtooth) if needed

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Power stage

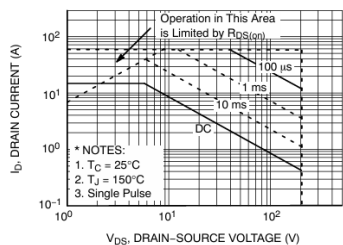
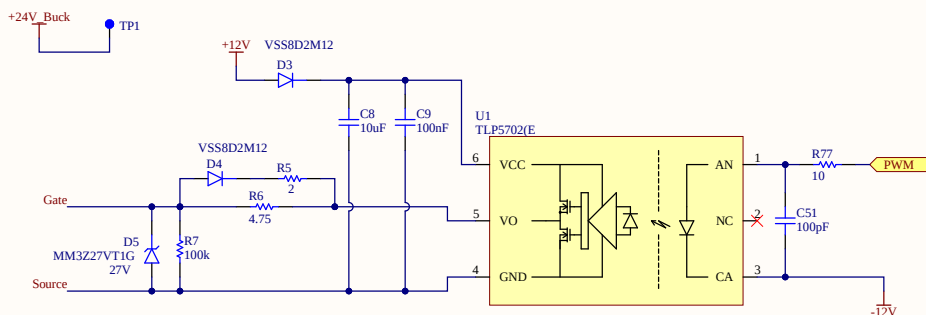
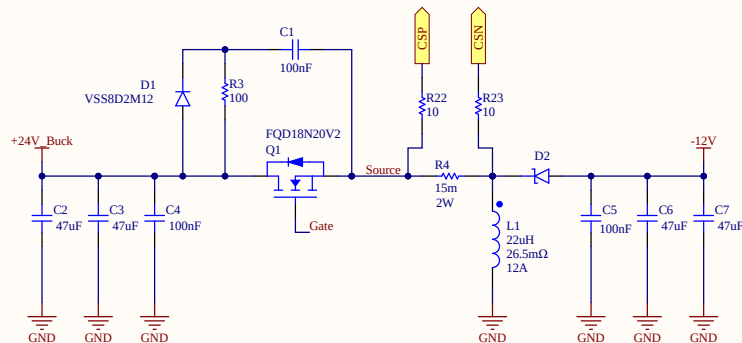


Figure 9. Maximum Safe Operating Area

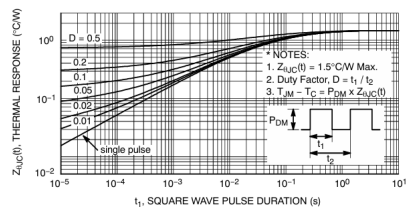


Figure 11. Transient Thermal Response Curve

Latch-off / hiccup OCP:
Q1 hits 11A once, OCP trips, gate goes off, converter stops (or waits to retry). The 396W linear event happens once → single pulse is correct.
Cycle-by-cycle OCP:
Q1 hits 11A, gets turned off for that cycle, then turns back on next cycle and hits 11A again... repeating every switching period for several ms during a sustained overload. Now the high-power linear crossing repeats every cycle → this is a pulse train, and you should use the duty-cycle curve, NOT single pulse.

FQD18N20V2 (Q1) — SOA CALCULATION

Package: DPAK | 200V | 15A | Rth(j-c)=1.5°C/W | TJmax=150°C
TA = 40°C (max ambient)
SOA points (Fig.9 @ VDS=36V, 25°C):
t1 = 1ms → ISOA = 20A
t2 = 10ms → ISOA = 7A
POWER-LAW FIT:
SOA(t) = a × t^m (t in seconds)
m = ln(20/7)/ln(1ms/10ms) = 1.050/-2.303 = -0.456
a = 20(1ms)^{-0.456} = 0.8575
SOA(t) = 0.8575 × t^{-0.456}

CASE 1 — NORMAL SWITCHING

- ON-state Vds in RDSon region, SOA-safe (low Vds)
- Switching edges: cross active region but last ~20ms << 10μs → far inside SOA pulse limit
- ID_peak = 5.31A << SOA(10μs) → far inside SOA ✓

CASE 2 — STARTUP (OCP clamp = 11A)

Vds = Vin + |Vout| = 36V
P_pulse = Vds × I_OCP = 36 × 11 = 396W
Zth(j-c) read from Fig.11 SINGLE-PULSE curve:
t1=1ms → Zth=0.17 t2=2ms → Zth=0.25 t3=3ms → Zth=0.30 (arbitrary sweep points to locate the current/pulse limit)

TJ = TA + Zth(t) × P_pulse
ISOA_derated = SOA(t) × (TJmax - TJ) / (TJmax - 25)

t=0.64ms: TJ=40+0.136×396=94°C
SOA25=0.8575×(0.64ms)^{-0.456}=24.5A
der=24.5×(150-94)/125=11.0A = 11A ◀ LIMIT
t=1ms: TJ=40+0.17×396=108°C
SOA25=20.0A, der=20.0×(150-108)/125=6.8A <11A ✗

CHECK: I_OCP=11A vs ISOA_derated
t≤0.64ms: 11A ≤ ISOA_derated ✓ PASS
t>0.64ms: 11A > ISOA_derated ✗ FAIL

RESULT

Max safe linear pulse at 11A/36V (TA=40°C) = 0.64ms
→ OCP must turn Q1 OFF within 0.64ms of reaching 11A.
Cycle-by-cycle: one period = 3.3μs @300kHz, far inside the 0.64ms safe window ✓

ANALYSIS

Vin=24V | Vout=-12V Iout=3A | fs=300kHz | D=0.333 | eff=88%

INDUCTOR — SRP1245A-220M (22μH)

IL_avg = Iin + Iout = 1.705 + 3 = 4.705A
ΔIL = Vin × D / (fs × L) = 1.21A
IL_peak = IL_avg + ΔIL/2 = 5.31A
IL_ripple_ratio = 1.21/4.705 = 25.7%
L_crit = (D)² × R / 2fs = 2.97μH
CCM confirmed: L=22μH >> Lcrit=2.97μH ✓

MOSFET — FQD18N20V2 (200 V, 15 A, 140 mΩ)

Vds_stress = Vin + Vout = 24+12 = 36V
Vds_rated = 36×1.2 = 43.2V → 600V ✓ (20% derating)
Id_rated ≥ IL_peak = 5.31A → 15A ✓
ID_rms = IL_avg × √D = 2.72A
P_cond = ID_rms² × Rds(on) × 1.3 = 2.72² × 0.637 = 4.71W (30% derating)
P_sw = 0.5 × Vds × IL_peak × (tr + tf) × fs = 0.5 × 36 × 5.31 × 16 × 300k = 0.46W
P_total = 4.71 + 0.57 = 5.12W

DIODE — STPS10L60S (60V, 10A, Vf=0.4V, DPAK)

Vr_stress = Vin + Vout = 36V
Vr_rated = 36×1.2 = 43.2V → 60V ✓
ID_avg = Iout = 3A
ID_peak = IL_peak = 5.31A
P_diode = Vf × Iout × (1-D) = 0.4 × 3 × 0.667 = 0.80W

OUTPUT CAPACITOR — KRM55WR71H476MH13K ×2 (47μF, 50V, X7R)

C_min = Iout × D / (fs × ΔVout) = 27.7μF
C_nominal = 2 × 47 = 94μF
C_derated @ 12V = 2 × 39.6 = 79.2μF > 27.7μF ✓
ΔVout_target = 1% × 12 = 120mV
ESR_max = 0.5 × ΔVout / ΔIL = 49.6mΩ
(the ESR can contribute max by a 50% to the total allowed ripple)
ESR_eff = 4mΩ/2 = 2mΩ << 49.6mΩ ✓
IC_rms = Iout × √(D/D') = 2.12A
IC_rms/cap = 2.12/2 = 1.06A ✓

INPUT CAPACITOR — KRM55WR71H476MH13K ×2 (47μF, 50V, X7R)

C_min = Iout × D / (fs × ΔVin × D') = 20.8μF
C_nominal = 2 × 47 = 94μF
C_derated @ 24V = 2 × 26.6μF = 53.2μF > 20.8μF ✓
ΔVin_target = 1% × 24 = 240mV
ESR_max = 0.5 × ΔVin / IL_peak = 22.6mΩ
ICin_rms = IL × √(D × D') = 2.22A ✓
Vrated = 32 × 1.5 = 48V → 50V ✓
Cin impedance Zcin = 1 / (2π × fs × Cin) + ESRcin = 0.413Ω
Cdamp impedance Zcdamp = 1 / (2π × fs × Cdamp) + ESRdamp = 0.013Ω
IRMS Total Input Caps (Cin + Cdamp) = IL × √(D × D') = 2.22A
IRMS Cdamp = Zcdamp / (Zcdamp + Zcin) = 2.16A ✓

POWER BUDGET

P_mosfet = 5.17W
P_diode = 0.80W
P_total = 6.08W
η_estimated = Pout / (Pout + Ploss) = 36 / (36 + 6.08) = 85.6%

MOSFET: FQD18N20V2

Driver: TLP5702(E)

VDRV = 24 V

Method 1 – Total Gate Charge

Qg = 20 nC

Idrv,max = 2.5 A

tr = Qg / Idrv,max = 8 ns

tf = Qg / Idrv,max = 8 ns

Method 2 – Miller Charge :

Qgd ≈ 7 nC

Vplateau ≈ 7.8 V

Ig,on = min[(VDRV - Vplateau)/Rgon, Idrv,max]

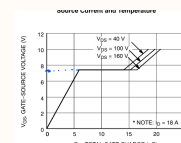
Ig,off = min[(VDRV - Vplateau)/Rgoff, Idrv,max]

tr = Qgd / Ig,on ≈ 2.8 ns

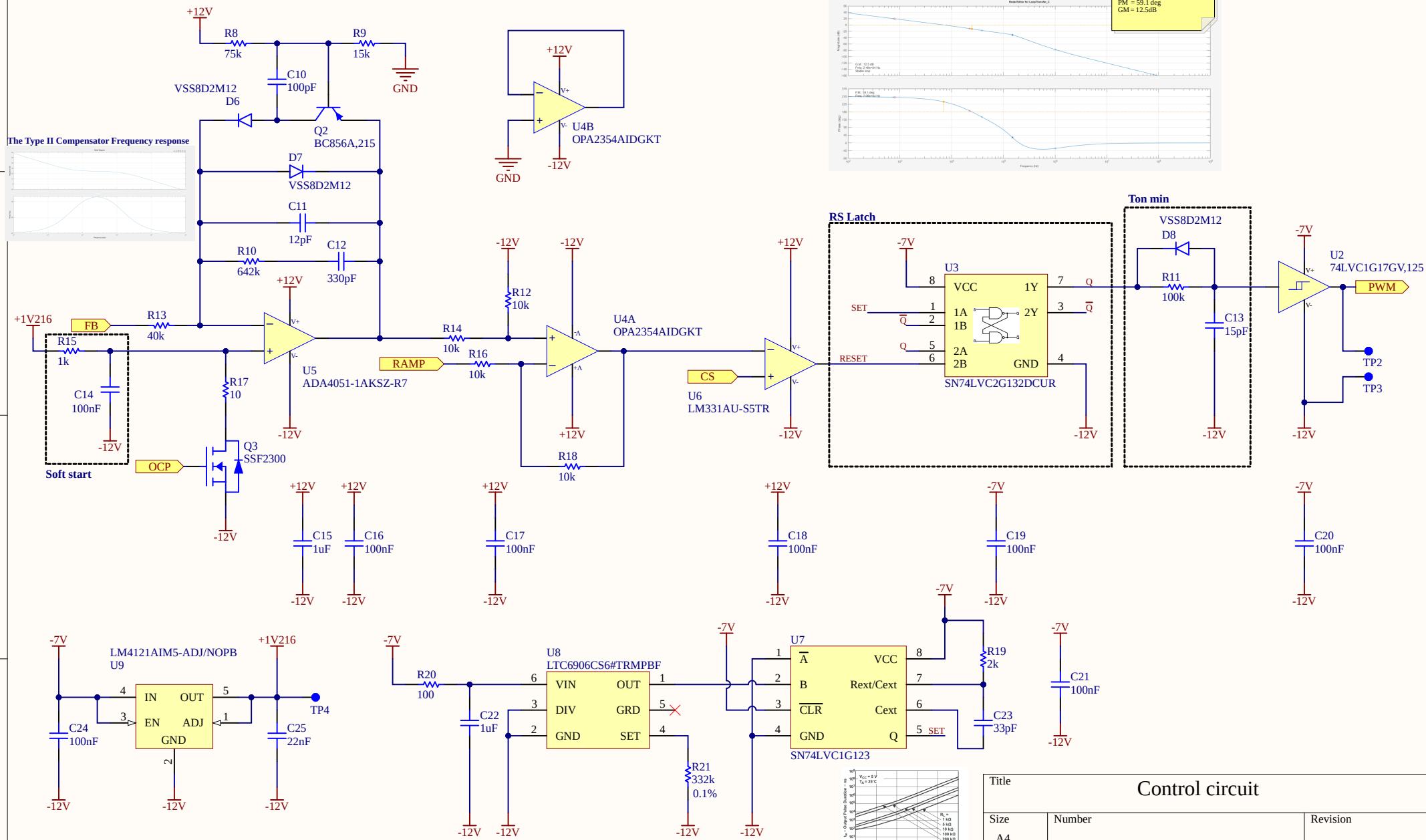
tf = Qgd / Ig,off ≈ 2.8 ns

Switching Loss:

Psw = 0.5 × VDS × ID × (tr + tf) × fs

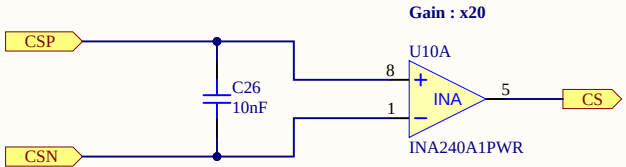


Control circuit

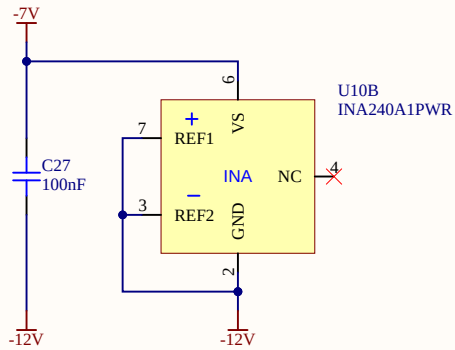


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Current sense



^A Total current gain = 0.015 x 20 = 0.3



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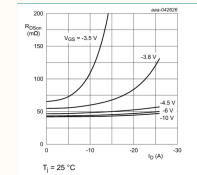


Fig. 7. Drain-source on-state resistance as a function of drain current; typical values

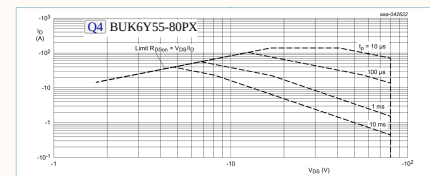
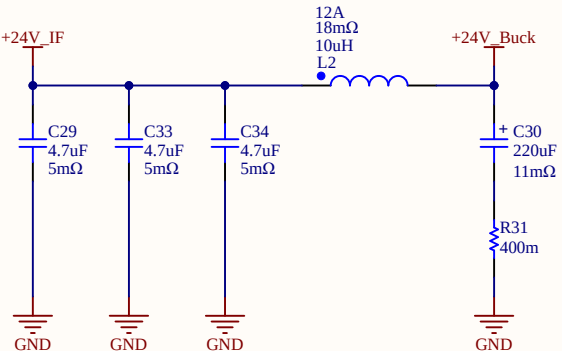
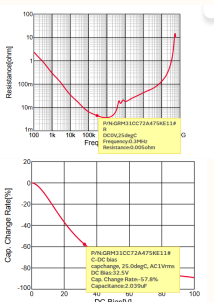
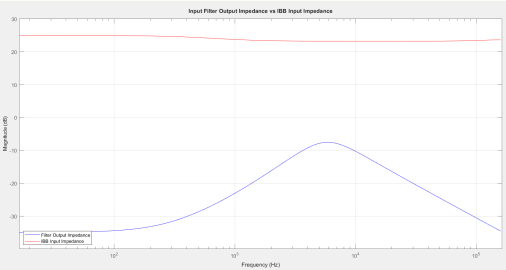


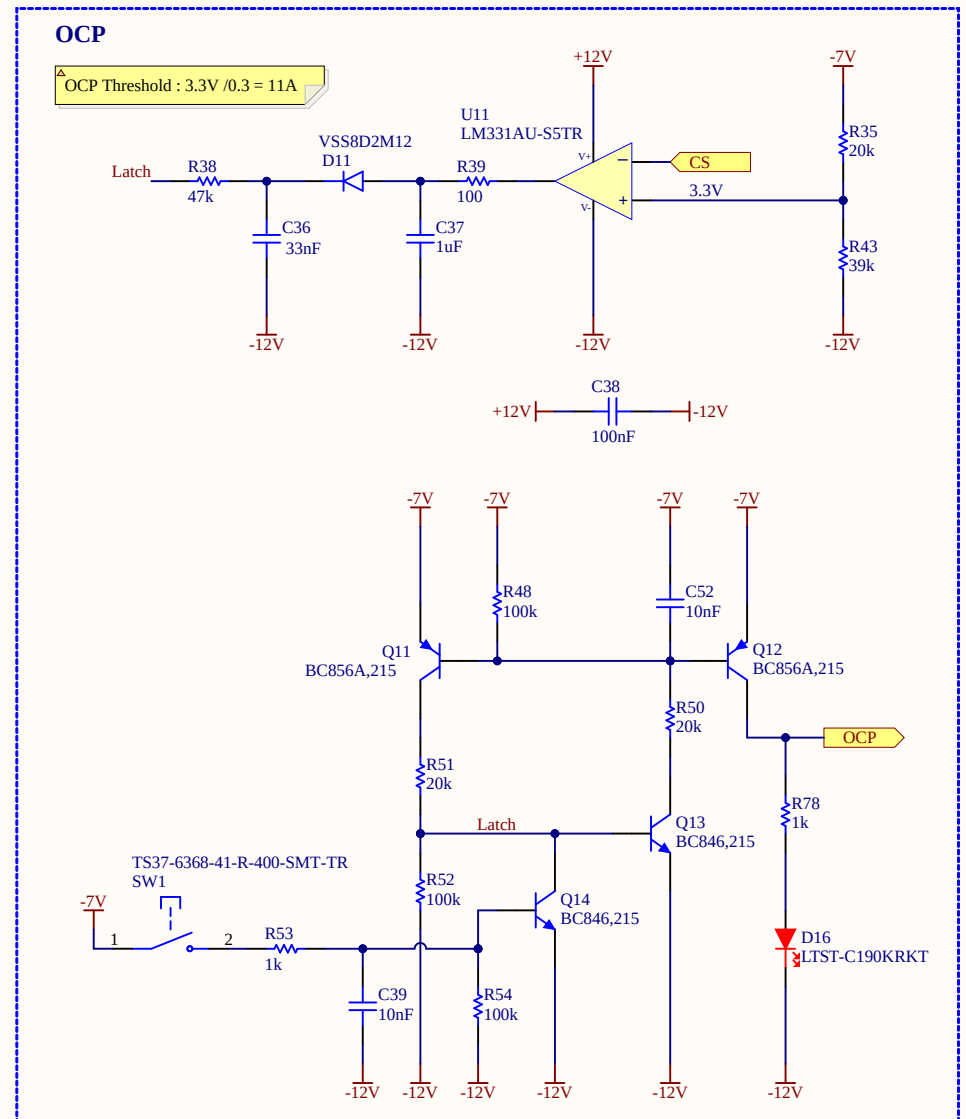
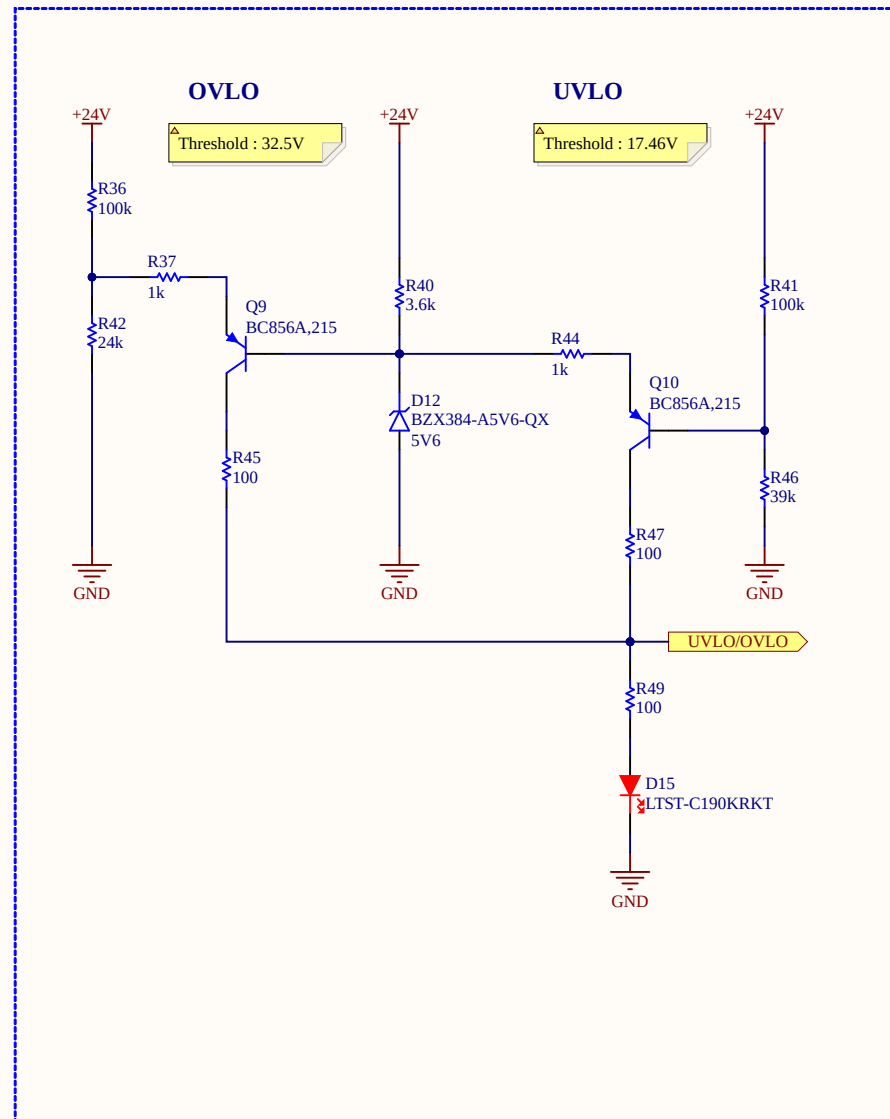
Fig. 3. Safe operating area; junction to mounting base; continuous and peak drain currents as a function of drain-source voltage

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^ INPUT EMI FILTER : Design Notes		
Standard: EN 55032:2015 Class B — Conducted Emissions		
Limit @ 300kHz = 56 dBμV (quasi-peak)		
Frequency range: 150kHz – 30MHz		
SPECIFICATIONS:		
Vin = 24V Vout = -12V Iout = 3A fs = 300kHz D = 0.333 Cin = 53.2uF (derated @24V)		
STEP 1 : NOISE LEVEL ESTIMATE		
First harmonic amplitude:		
I1 = (2×IL×sin(π×D)) / π = (2×4.705×sin(π×0.333)) / π = 2.60A		
Noise voltage across Cin at fs:		
Vnoise = I1 / (2π×fs×Cin) = 2.60 / (2π×300k×53.2u) = 25.9mV = 25.9×10³ μV		
Estimated noise level:		
Noise dBμV = 20log(25.9×10³) = 88.3 dBμV		
Required attenuation:		
Att = Noise dBμV - Vmax = 88.3 - 56 = 32.3 dB		
Adding 10dB safety margin:		
Att_target = 32.3 + 10 = 42.3 dB		
STEP 2 : INDUCTOR SELECTION		
Lf = 10uH → Selected: 10uH, 12A, SMD		
STEP 3 : CAPACITOR SELECTION		
Resonance one decade below fs:		
Cf = Cin / (Cin×Lf×(2π×fs/10)² - 1) = 53.2u / (53.2u×10u×(2π×30k)² - 1) = 3μF		
Required attenuation:		
Cf= 1/Lf × (10^(Att/40)/(2π×fs))² = 5μF		
Select higher value: Cf = max(5μF,3μF) = 5μF→ 3x 4.7μF selected derated to 2uF @Vin,max = 32.5V		
NOTE: Cf is a SEPARATE dedicated filter capacitor distinct from Cin.		
fc_filter = 1/(2π×√(Lf×Cf)) = 1/(2π×√(10u×6μF)) = 20.5kHz		
Attenuation verification:		
A = 40×log(fs/fc_filter) = 40×log(300k/20.5k) = 46.6dB >> 42.3dB ✓		
STEP 4 : DAMPING CAPACITOR		
Cd ≥ 4×Cin = 4×53.2u = 212.8uF → 220uF selected		
Required ESRd:		
ESRd ≤ √(Lf/Cin) = √(10u/53.2u) = 0.434Ω		
Damping resistor power:		
P_Rdamp ≈ 0.5×Δlin²×Rdamp = 0.5×(1.21)²×1.5 = 0.8W		
Selected Cd: 220uF 35V polymer electrolytic ESR_polymer ≈ 11mΩ < 434mΩ ✓		
NOTE: Polymer ESR too low, added series resistor Rd = 0.4Ω in series with Cd to meet ESRd target		
Voltage rating: Vin_max×1.25 = 32×1.25 = 40V → 50V ✓		
Cin impedance Zcin = 1/(2π×fs×Cin) + ESRcin = 0.413Ω		
Cdamp impedance Zcdamp = 1/(2π×fs×Cdamp) + ESRdamp = 0.013Ω		
IRMS Total Input Caps (Cin +Cdamp) = IL×√(D×D') = 2.22A		
IRMS Cdamp (current divider) = Zcin / (Zcdamp +Zcin) = 0.062A✓		
STEP 4 : MIDDLEBROOK CRITERION		
Zin_conv = Vin²×η/Pout = 576×0.88/36 = 14.1Ω		
Full frequency spectrum verified using matlab: see Plot		
EMC STANDARD : EN 55032:2015		
EN 55032 "Electromagnetic Compatibility of Multimedia Equipment" . It defines conducted emission limits on the power port for Class B equipment (residential environment):		
150kHz – 500kHz : 56–66 dBμV (quasi-peak)		
46–56 dBμV (average)		
500kHz – 30MHz : 56 dBμV (quasi-peak)		
46 dBμV (average)		
Worst case for this design at fs=300kHz: 56 dBμV QP.		
Reference : Conducted EMI Characteristics And Mitigation Technique		
https://www.ti.com/lit/an/snva489c/snva489c.pdf		

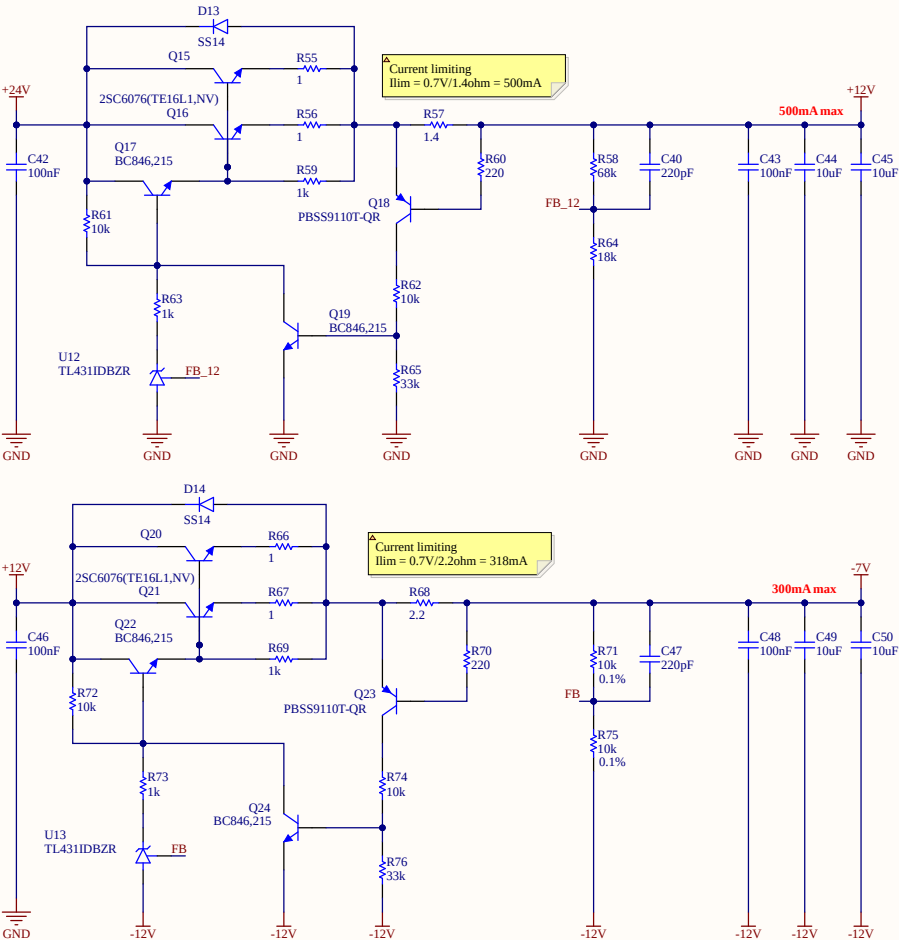
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Protections



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Linear regulators



To obtain the maximum possible current gain using Darlington pair it's important to ensure that the driver transistor has enough collector current to ensure acceptable hFE. When run at very low collector current, most transistors will have a gain that's well below the quoted figure in the datasheet. The base-emitter resistor (R59 and R69) is sized to ensure that the drive transistor has a collector current that's above the minimum. The resistor also helps to ensure a faster turn-off and minimises leakage current.

2SC6076 POWER PASS TRANSISTORS — SOA & THERMAL VERIFICATION

DATASHEET (Toshiba 2SC6076):
VCEO = 80V | VCB0 = 160V | IC = 3A DC / 5A pulse Pc = 10W @ Tc=25°C | Tj_max = 150°C
VCE(sat) = 0.5V max @ IC=1A | hFE = 100-450
Thermal resistances (large copper):
Rth(c-a) = 21.2 °C/W
Rth(j-c) = 2 °C/W
Linear-regulator operation = CONTINUOUS → use DC, SOA.

RAIL 1 — +12V REGULATOR (Q15 // Q16)

OPERATING POINT (per device, current shared):

Vce = Vin - Vout = 24 - 12 = 12V
Ic_total = Ilim = 500mA
Ic_perdevice = 500mA / 2 = 250mA

POWER DISSIPATION (per device):

Pd = Vce × Ic_perdevice = 12 × 0.25 = 3.0W (< Pc=10W @ Tc=25°C ✓)

SOA CHECK (DC operation, continuous):

Operating point: Vce=12V, Ic=250mA

DC SOA curve @ Vce=12V:

IC_SOA(DC, 12V, 25°C) = 0.8A per device

THERMAL:

Rth(c-a) = 21.2 °C/W | Rth(j-c) = 2 °C/W

Tc = TA + Rth(c-a) × Pd = 40 + 21.2 × 3.0 = 103.6 °C

Tj = Tc + Rth(j-c) × Pd = 103.6 + 2 × 3.0 = 109.6 °C ✓ < 150°C

Derated SOA to Tc:

Ic_SOA(DC, Tc) = 0.8 × (150-Tc)/(150-25)

= 0.8 × (150-103.6)/(150-25)

= 0.297A

CHECK: Ic_perdevice = 0.25A < Ic_SOA_derated = 0.297A ✓

SOA margin = 0.297/0.25 = 1.19×

RAIL 2 — -7V REGULATOR (Q20 // Q21)

OPERATING POINT (per device, current shared):

Vce = Vin - Vout = 12 - (-7) = 19V

Ic_total = Ilim = 318mA

Ic_perdevice = 318mA / 2 = 159mA

POWER DISSIPATION (per device):

Pd = Vce × Ic_perdevice = 19 × 0.159 = 3.021W (< Pc=10W ✓)

SOA CHECK (DC operation, continuous):

Operating point: Vce=5V, Ic=159mA

DC SOA curve @ Vce=5V:

IC_SOA(DC, 5V, 25°C) = 2.0A per device

THERMAL:

Rth(c-a) = 21.2 °C/W | Rth(j-c) = 2 °C/W

Tc = TA + Rth(c-a) × Pd = 40 + 21.2 × 3.021 = 104.0 °C

Tj = Tc + Rth(j-c) × Pd = 104.0 + 2 × 3.021 = 110.0 °C ✓ < 150°C

Derated SOA to Tc:

Ic_SOA(DC, Tc) = 2.0 × (150-Tc)/(150-25)

= 2.0 × 46.0/125

= 0.736A

CHECK: Ic_perdevice = 0.159A < Ic_SOA_derated = 0.736A ✓

SOA margin = 0.736/0.159 = 4.63×

EMITTER BALLAST — CURRENT SHARING

RE = 1Ω (R55/R56 and R66/R67)

Purpose: forces current sharing between parallel devices, prevents thermal runaway from VBE mismatch.

Voltage drop across RE:

Rail1: VRE = 0.25A × 1 = 250mV

Rail2: VRE = 0.159A × 1 = 159mV

VRE ≥ 5×ΔVBE rule (ΔVBE=10mV typical mismatch (between the 2 trnasistors in //):

5×10mV = 50mV → Rail1 (250mV) ✓ / Rail2 (159mV) marginal

PRE: Rail1 = 0.25×1 = 250mW → 0.500W rated ✓

RESULT

RAIL 1 (+12V): Vce=12V, Pd=3.0W/dev, Tj=109.6°C ✓ PASS

RAIL 2 (-7V): Vce=19V, Pd=3.02W/dev, Tj=110.0°C ✓ PASS

Both parallel pairs operate within SOA and thermal

limits well below 150°C max.

Emitter ballast RE = 1Ω (single value both rails):

+12V: V_RE=250mV = 5.0×ΔVBE ✓

-7V: V_RE=159mV = 3.2×ΔVBE (acceptable)

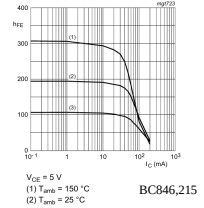
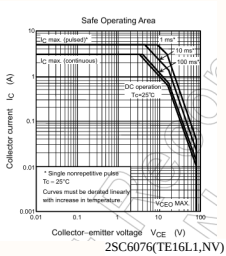


Fig. 2. Group A: DC current gain as a function of collector current; typical values

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